

For full credit explain your reasoning, showing all relevant work.

Exercise 1. Let $G = \langle a \rangle$ be a cyclic group of order 20 generated by an element a .

a) What are all possible generators of G ? That is, what are elements a^k such that $G = \langle a^k \rangle$?

b) Let H be a subgroup of G of order 10. What are all possible generators of H ? That is, what are elements a^k such that $H = \langle a^k \rangle$?

Exercise 2. a) Let G be a cyclic group which has exactly 3 different subgroups: the whole group G , the trivial subgroup $\{e\}$ and a subgroup H of order 11. What is the order of G ? Justify your answer.

b) Let G be a cyclic group which has as the only subgroups the whole group G , and groups of orders 1, 2, 4, 7, 14. What is the order of G ? Justify your answer.

Exercise 3. Let $\mathbb{Q}$ be the group of rational numbers with addition. Show that $\mathbb{Q}$ is not a cyclic group.

PRACTICE PROBLEMS

Exercises below are for practice only - do not turn them in for grading.

Practice Exercise 1. Find all elements of order 9 in the group $\mathbb{Z}_{900}$. Explain how you know that these are all elements of order 9 in this group.

Practice Exercise 2. Let $\mathbb{Q}^+$ denote the group of positive rational numbers with multiplication. Show that $\mathbb{Q}^+$ is not a cyclic group.

Practice Exercise 3. For each integer $n \geq 1$ give an example of a group G which has exactly n different subgroups (including the whole group and the trivial subgroup).

Practice Exercise 4. Show that if G is an infinite group then it has infinitely many different subgroups.